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4U 1626–67 as seen by *Suzaku* before and after the 2008 torque reversal

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ABSTRACT

Aims. The accretion-powered pulsar 4U 1626–67 experienced a new torque reversal at the beginning of 2008, after about 18 years of steadily spinning down. The main goal of the present work is to study this recent torque reversal that occurred in February 2008.
Methods. We present a spectral analysis of this source using two pointed observations performed by *Suzaku* in March 2006 and in September 2010.
Results. We confirm with *Suzaku* the presence of a strong emission-line complex centered on 1 keV, with the strongest line being the hydrogen-like Ne Lyα at 1.025(3) keV. We were able to resolve this complex with up to seven emission lines. A dramatic increase in the intensity of the Ne Lyα line after the 2008 torque reversal occurred, with the equivalent width of this line reaching almost the same value measured by ASCA in 1993. We also report on the detection of a cyclotron line feature centered at ∼37 keV. In spite of the increase in the X-ray luminosity (0.5−100 keV) of a factor of ∼2.8 that occurred between these two observations, no significant change in the energy of the cyclotron line feature was observed. However, the intensity of the ∼1 keV line complex increased by an overall factor of ∼8.
Conclusions. Our results favor a scenario in which the neutron star in 4U 1626–67 accretes material from a geometrically thin disk during both the spin-up and spin-down phases.

Key words. accretion, accretion disks – binaries: close – stars: neutron – pulsars: individual: 4U 1626–67 – X-rays: binaries

1. Introduction

The Roche-lobe filling binary system, 4U 1626–67, has matter from the companion flows from the L1 point into an accretion disk (Reynolds et al. 1997). This low mass X-ray binary (LMXB) consists of a 7.66 s X-ray pulsar accreting from an extremely low mass companion ((0.04 M$\odot$ for $i = 18°$, Levine et al. 1988). Although orbital motion has never been detected in the X-ray data, pulsed optical emission reprocessed on the surface of the secondary revealed the 42 min orbital period (Middleditch et al. 1981), confirmed by Chakrabarty (1998). Most likely the binary system contains a hydrogen-depleted secondary to reach such a short orbital period (Paczynski & Sienkiewicz 1981). The faint optical counterpart (KZ TrA, $V \sim 17.5$) has a strong UV excess and high optical pulse fraction (McClintock et al. 1977, 1980).

The sign of the torque in 4U 1626–67 has reversed on only two occasions since its discovery by *Uhuru* (Giacconi et al. 1972). Monitoring of the source by the Burst And Transient Source Experiment (BATSE) on board the *Compton* Gamma Ray Observatory (CGRO) starting in April 1991, found the pulsar spinning down, implying a changed sign in the accretion torque (Wilson et al. 1993; Bildsten et al. 1994). It was estimated that the reversal occurred in mid-1990. More recently, our daily monitoring of 4U 1626–67 with *Fermi*/GBM starting in August 2008 found the pulsar again spinning-up rather than spinning-down. *Swift*/BAT observations allowed us to cover the evolution of this second torque reversal. The transition took place at around MJD 54 500 (2008 Feb. 04) and lasted approximately 150 days (Camero-Arranz et al. 2010).

A 48 mHz quasi-periodic oscillation (QPO) has been detected in the X-ray emission (Shinoda et al. 1990; Kommers et al. 1998). Orlandini et al. (1998) inferred a neutron star magnetic field in the range of $(2.4–6.3) \times 10^{12}$ G. To compute this magnetic field range, a source distance of 5−13 kpc was assumed (Chakrabarty 1998, and references therein). A ∼37 keV absorption cyclotron feature has been found in the 0.1−200 keV *BeppoSAX* spectrum (Orlandini et al. 1998). A more detailed study of this feature with *Suzaku* can be found in Iwakiri et al. (2012). The X-ray broad-band continuum was fitted with low-energy absorption, a blackbody, a power law and a high energy cutoff.

Angelini et al. (1995), using the ASCA observatory, discovered for the first time a strong emission-line complex centered on 1 keV in the X-ray spectrum of 4U 1626–67. The strongest line was identified as hydrogen-like Ne Lyα at 1.008 keV. The strength of the neon emission compared to the expected iron L complex implies a large neon overabundance.

We present a spectral analysis of 4U 1626–67 with *Suzaku*, using two observations from March 2006 and September 2010.

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朱雀在 2008 年扭矩反转前后观测到的 4U1626–67

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摘要

目标。吸积驱动的脉冲星 4U 1626–67 在经历了约 18 年的持续减速后，于 2008 年初经历了新一轮的扭矩反转。本研究的主要目标是探讨 2008 年 2 月发生的这一近期扭矩反转现象。

方法。我们利用*Suzaku*于 2006 年 3 月和 2010 年 9 月进行的两次定点观测，对该源进行了频谱分析。

结果。我们通过与*Suzaku*的观测确认，在 1 keV 附近存在一个强烈的发射线复合体，其中最强的谱线是位于 1.025(3) keV 的类氢 Ne Lyα 线。我们成功解析出该复合体中的多达七条发射线。2008 年扭矩反转后，Ne Lyα 线的强度显著增加，其等效宽度几乎达到了 ASCA 在 1993 年测得的相同数值。我们还报告了在一个中心位于 ~37 keV 的回旋线特征被探测到的结果。尽管在这两次观测之间，X 射线光度（0.5_100 keV）增加了约 2.8 倍，但回旋线特征的能量并未发生显著变化。然而，~1 keV 线复合体的强度整体增加了约 8 倍。

结论。我们的研究结果支持这样一种情景：4U 1626–67 中的中子星在自转加速阶段和自转减速阶段均从几何薄盘中吸积物质。

关键词。吸积，吸积盘 – 双星：近距离 – 恒星：中子星 – 脉冲星：单个：4U 1626–67 – X 射线：双星

1. 引言

Roche瓣填充双星系统4U 1626–67，其伴星物质从L1点流入吸积盘（Reynolds 等人 1997）。这个低质量X射线双星（LMXB）由一个7.66秒X射线脉冲星和一个极低质量伴星组成（当i=18 °时，伴星质量为0.04倍太阳质量，Levine等人 1988）。尽管在X射线数据中从未检测到轨道运动，但次级星表面再处理的脉冲光学辐射揭示了42分钟的轨道周期（Middleditch 等人 1981），这一结果得到了Chakrabarty（1998）的证实。最可能的情况是，该双星系统包含一个氢元素贫化的次级星，从而实现了如此短的轨道周期（Paczynski & Sienkiewicz 1981）。其微弱的光学对应体（KZ TrA，V~ 17.5）具有强烈的紫外超量和较高的光学脉冲比例（McClintock等人1977，1980）。

自*乌胡鲁*发现4U 1626–67的扭矩符号以来，该符号仅在两次发生反转（贾科尼等人1972）。自1991年4月起，由搭载在*康普顿*伽马射线天文台（CGRO）上的爆发与瞬变源实验（BATSE）对源进行监测，发现该脉冲星正在减速旋转，这表明吸积扭矩的符号发生了改变（威尔逊等人1993；比尔德斯滕等人1994）。据估计，这种反转发生在1990年中期。最近，我们自2008年8月起利用*Fermi*/ GBM 对4U 1626–67进行日常监测，发现该脉冲星再次开始加速旋转，而非减速旋转。*Swift*/BAT的观测使我们能够追踪这一第二次扭矩反转的演化过程。转变发生在 MJD 54 500左右（2008年2月4日），持续约150天（Camero-Arranz等2010）。

在X射线辐射中探测到了一种48 mHz的准周期性振荡（QPO）（Shinoda 等人 1990；Kommers 等人 1994）。Orlandini等人（1998）推断中子星磁场强度在(2.4_6.3) × 10^{12} G范围内。为计算此磁场范围，假设源距离为5_13 kpc（Chakrabarty 1999，及其参考文献）。在0.1_200 keV的*BeppoSAX*光谱中发现了约37 keV的吸收回旋共振特征（Orlandini等人1998）。关于该特征的更详细研究可参阅Iwakiri 等人（2012）利用*Suzaku*进行的分析。X射线宽带连续谱通过低能吸收、黑体辐射、幂律和高能截止进行了拟合。

Angelini 等人（1995 年）利用 ASCA 天文台，首次在 4 U 1626–67 的 X 射线光谱中发现了以 1 keV 为中心的强发射线复合体。最强的谱线被鉴定为位于 1.008 keV 的类氢氖 Lyα 线。氖发射线的强度与预期的铁 L 复合体相比，表明氖的丰度较高。

我们利用*Suzaku*对 4U 1626–67 进行了光谱分析，采用了 2006 年 3 月和 2010 年 9 月的两次观测数据。

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This allows us to better characterize the recent torque reversal that occurred in February 2008.

2. Observations and analysis

The instruments on *Suzaku* (Mitsuda et al. 2007) are the X-ray Imaging Spectrometer (XIS; Koyama et al. 2007) CCD detector covering the 0.3−10 keV band, and the Hard X-ray Detector (HXD; Takahashi et al. 2007) comprised of the PIN diode detector (PIN) covering the 10−70 keV band and the gadolinium silicate crystal detector (GSO) covering the 60−600 keV band. The XIS has four separate detectors, XIS 0–3, with XIS 1 being a back-illuminated CCD and the others frontside. XIS 2 was lost due to a micrometeor hit in late 2006.

The first *Suzaku* 4U 1626–67 observation occurred in 2006 March (MJD 53 797) and thus before the torque reversal in 2008. During this observation the spacecraft aimpoint was optimized for the XIS detectors, and *Suzaku* was run in a 1/8 window mode with 1 s integration time, an effective area of 128 × 1024 pixels and an effective exposure time of ∼72.2 ks. The second observation in 2010 September (MJD 55 444) was also centered on the XIS aimpoint but run in a 1/4 window mode, with 2 s integration time, an effective area of 256 × 1024 pixels and an effective exposure time of ∼14.01 ks. For this observation only 3 XIS detectors were available, as we discussed in the previous paragraph. The *Suzaku* data were reduced with tools from the HEASOFT v6.11 package (Arnaud 1996) and CALDB release 20110804. Following the *Suzaku* ABC guide[1], the first step of our analysis was to reprocess the data by running the FTOOL *aepipeline* for the XIS, the PIN and the GSO. The pipeline performs calibration as well as data screening.

Due to thermal distorsion of the *Suzaku* spacecraft the optical bench shows wobbling. The source's PSF image is not centered on a fixed position on the XIS CCDs but appears blurred. As part of the standard data reduction this effect is partially corrected by adjusting the spacecraft attitude file. The attitude file can be further improved by applying the `FTOOL aeattcor2`: based on the algorithm described by Nowak et al. (2011) the mean event positions as a function of time are calculated and used to further adjust the attitude. While we obtain sharp images for our first *Suzaku* observation, the second one suffers from additional attitude variability that can be present in *Suzaku* observations since a change in the attitude control system on 2009 Dec. 18. In this case the `FTOOL` cannot fully correct the characteristic "double images". According to the documentation[2], however, and since conservative pileup exclusion regions were applied (see below), the XIS spectra are not significantly affected.

We used the `FTOOL pileest` to estimate the amount of pileup in the XIS images and disregarded circular regions with >8% pileup fraction during the spectral extraction. This is a careful approach since due to the asymmetric shape of the PSF only few pixels outside of the exclusion region reach pileup fractions above 2–4%. For the first observation the pileup fraction remained below this limit in the entire source extraction region (a circle of ∼70″ radius centered on the source image). In the second observation pileup fractions of up to 15% were reached in the center of the source image so the extraction region consisted of a circle of ∼130″ radius minus a central circular exclusion region with a radius of ∼35″.

The primary tool for extracting data products (spectra, lightcurves) from XIS data is *xselect*, which is part of the general HEAsoft distribution. Events in the XIS detectors were in the 3 × 3 and 5 × 5 editing modes. With *xselect* we applied to the cleaned events filters with the good times intervals avoiding telemetry saturation. Source and background regions were created from the image. The filtered events were then used to create source and background spectra for the desired extraction regions. Products from the two editing modes were added. The XIS spectra were grouped close to the energy resolution of the detectors (see Nowak et al. 2011). We created response matrices and effective area files with the *xisrmfgen* and *xissimarfgen* tools, respectively. The calibration of the XIS is uncertain around the Si K edge, thus events in the 1.8−2.0 keV range were excluded in the fitting. For more information about this issue we refer the reader to the "XIS Calibration around the Si K edge" web page[3] (see also Iwakiri et al. 2012; Nowak et al. 2011; Suchy et al. 2011, for different treatments on this matter).

For the PIN, we extracted spectra from the cleaned event files using *hxdpinxbpi*. This tool produces the dead time corrected PIN source spectrum as well as the PIN background spectrum. The non X-ray background events provided by the *Suzaku* team can be downloaded from the HEASARC archive[4]. The PIN spectra were grouped to have a signal-to-noise ratio ≳10 in each energy bin, and we considered spectra between 14−70 keV. We point out that due to the changes in instrumental settings the PIN response matrices, from the CALDB database, are different depending on the epoch of the observation.

The GSO spectra were created starting with the clean event files using the *hxdgsoxbpi* tool. The background events were downloaded from HEASARC[5]. Response files were taken from the CALDB database and the GSO correction ARF file was provided by the HXD instrument team[6]. The distributed GSO Non X-ray Background is grouped, so GSO spectra have to be grouped accordingly, using the spectral binning given by the file *gsobgd64bins.dat* available at HEASARC[7]. The GSO spectrum was restricted to the 50−70 keV range for the first observation, and to the 50−100 keV for the second observation.

3. Results

The ∼0.5−100 keV broad-band spectrum obtained using XIS (a combination of detectors 0 and 3), PIN and GSO were fitted in XSPEC 12.7.0. We used a canonical model which allowed us to compare our spectral study with previous works (Angelini et al. 1995; Pravdo et al. 1979; Orlandini et al. 1998; Krauss et al. 2007; Jain et al. 2010, among others) and update the long-term X-ray flux history of 4U 1626–67 relative to the flux measured by HEAO 1 (Chakrabarty et al. 1997; Krauss et al. 2007; Camero-Arranz et al. 2010).

The continuum model fitted to both observations includes low-energy absorption, a blackbody component, and a power law with a high-energy cutoff at ∼20 keV

[1] `http://heasarc.nasa.gov/docs/suzaku/analysis/abc`
[2] `ftp://legacy.gsfc.nasa.gov/suzaku/doc/general/suzakumemo-2010-05.pdf`
[3] `http://heasarc.gsfc.nasa.gov/docs/suzaku/analysis/sical.html`
[4] `ftp://legacy.gsfc.nasa.gov/suzaku/data/background/pinnxb_ver2.0_tuned`
[5] `ftp://legacy.gsfc.nasa.gov/suzaku/data/background/gsonxb_ver2.5`
[6] `http://www.astro.isas.ac.jp/suzaku/analysis/hxd/gsoarf/`
[7] `ftp://legacy.gsfc.nasa.gov/docs/suzaku/analysis/gsobgd64bins.dat`

这使我们能够更准确地描述2008年2月发生的近期扭矩反转现象。

2. 观察与分析

*Suzaku*上的仪器（Mitsuda 等人2007）包括覆盖 0.3–10 keV 能段的 X 射线成像光谱仪（XIS；Koyama 等人2005）CCD 探测器，以及由覆盖 10–70 keV 能段的 PIN 二极管探测器（PIN）和覆盖 60–600 keV 能段的钆硅酸盐晶体探测器（GSO）组成的硬 X 射线探测器（HXD；Takahashi 等人 2007）。XIS 拥有四个独立的探测器，即 XIS 0–3，其中 XIS 1 为背照式 CCD，其余为正面照式。XIS 2 因 2006 年末被微陨石撞击而损毁。

第一次*Suzaku* 4U 1626–67 观测发生在 2006 年 3 月（MJD 53 797），因此早于 2008 年的扭矩反转。在此次观测期间，航天器的目标点针对 XIS 探测器进行了优化，*Suzaku*以 1/8 窗口模式运行，积分时间为 1 秒，有效面积为 128 × 1024 像素，有效曝光时间为约 72.2 ks。第二次观测于 2010 年 9 月（MJD 55 444），同样以 XIS 目标点为中心，但以 1/4 窗口模式运行，积分时间为 2 秒，有效面积为 256 × 1024 像素，有效曝光时间为约 14.01 ks。由于如前文所述，此次观测仅可使用 3 个 XIS 探测器。*Suzaku* 数据使用 heasoft v6.11 软件包（Arnaud 1996）和 CALDB 20110804 版本中的工具进行了处理。遵循*Suzaku* ABC 指南[1]，我们分析的第一步是通过运行针对 XIS、PIN 和 GSO 的 FTOOL *aepipeline*对数据进行重新处理。该流程不仅执行校准，还进行数据筛选。

由于*朱雀号*航天器存在热畸变，光学平台会出现晃动现象。源的光斑点扩散函数图像并未以XIS CCD上的固定位置为中心，而是呈现模糊状态。作为标准数据还原的一部分，通过调整航天器姿态文件可部分校正此效应。若采用 FTOOL aeattcor2算法，可进一步改善姿态文件：基于Nowak等人（2011）所述的算法，计算事件平均位置随时间的变化函数，并用于进一步调整姿态。虽然我们在首次*朱雀号*观测中获得了清晰的图像，但第二次观测则受到额外的姿态变化影响；这种变化可能出现在*朱雀号*的观测中，因为姿态控制系统在2009年12月18日发生了变更。在这种情况下，FTOOL 无法完全校正特征性的“双像”现象。然而，根据文献[2]的说明，且由于采用了保守的堆积排除区域（见下文），XIS光谱并未受到显著影响。

我们使用 FTOOL pileest 来估算 XIS 图像中的堆积量，并在光谱提取过程中忽略堆积比例 >8% 的圆形区域。这是一种谨慎的做法，因为由于 PSF 的不对称形状，只有少数位于排除区域外的像素其堆积比例会超过 2_4%。在第一次观测中，整个源提取区域（以源图像为中心、半径约为 ~70″的圆形区域）的堆积比例始终低于这一限值。在第二次观测中，源图像中心的堆积比例最高达到 15%，因此提取区域由一个半径约为 ~130″的圆形区域组成，从中减去一个半径约为 ~35″的中央圆形排除区域。

从XIS数据中提取数据产品（能谱、光变曲线）的主要工具是*xselect*，它是HEAsoft通用软件套件的一部分。XIS 探测器中的事件采用3 × 3 and 5 × 5编辑模式。通过*xselect*，我们将经过清理的事件应用于过滤器，并设置合适的时间区间以避免遥测饱和。根据图像创建源区和背景区。随后，利用过滤后的事件为所需的提取区域生成源谱和背景谱。将两种编辑模式产生的结果合并。XIS能谱按探测器能量分辨率附近进行分组（参见Nowak等人2011）。我们分别使用xisrmfgen和xissimarfgen工具创建了响应矩阵和有效面积文件。XIS在Si K边附近的校准存在不确定性，因此将1.8_2.0 keV范围内的事件排除在拟合之外。有关此问题的更多信息，请参阅“XIS在Si K边附近的校准”网页[3]（另见Iwakiri等人2012；Nowak等人2011；Suchy 等人2011 年，针对该问题采取了不同的处理方式）。

对于 PIN，我们使用*hxdpinxbpi*从经过清理的事件文件中提取光谱。该工具可生成经死时间校正后的 PIN 源光谱以及 PIN 背景光谱。由*Suzaku*团队提供的非 X 射线背景事件可从 heasarc 存档[4]中下载。将 PIN 光谱进行分组，以获得一定的信噪比≥每个能量区间内为10个，我们考虑了14_70 keV之间的光谱。需要指出的是，由于仪器设置的变化，从 CALDB 数据库中获取的PIN响应矩阵会因观测时段的不同而有所差异。

GSO 能谱的生成始于使用*hxdgsoxbpi*工具处理干净的事件文件。背景事件从 heasarc [5]下载。响应文件取自 CALDB 数据库，GSO 校正 ARF 文件则由 HXD 仪器团队提供[6]。分布的 GSO 非 X 射线背景经过分组处理，因此 GSO 能谱也必须相应地进行分组，使用 heasarc [7]提供的文件*gsobgd64bins.dat*中给出的能谱分箱。首次观测时，GSO 能谱被限制在 50_70 keV 范围内；第二次观测时则限制在 50_100 keV 范围内。

3. 结果

使用XIS（探测器0和3的组合）、PIN和GSO获得的约0.5_100 keV宽带谱在 XSPEC 12.7.0中进行了拟合。我们采用了一个标准模型，使我们能够将光谱研究与先前的工作（Angelini等人1995；Pravdo等人1979；Orlandini等人1998；Krauss等人2007；Jain等人2010，以及其他研究）进行比较，并更新4U 1626–67的长期X射线流量历史，相对于 HEAO 1测得的流量（Chakrabarty等人1997；Krauss等人2007；Camero-Arranz等人2010）。

适用于两项观测数据的连续体模型包含低能吸收、黑体成分，以及一个在约 20 keV 处具有高能截断的幂律成分。

[1] http://heasarc.nasa.gov/docs/suzaku/analysis/abc[2] ftp://legacy.gsfc.nasa.gov/suzaku/doc/general/suzakumemo-2010-05.pdf

[3] http://heasarc.gsfc.nasa.gov/docs/suzaku/analysis/ sical.html

[4] ftp://legacy.gsfc.nasa.gov/suzaku/data/background/ pin-nxb_ver2.0_tuned

[5] ftp://legacy.gsfc.nasa.gov/suzaku/data/background/ gso-nxb_ver2.5

[6] http://www.astro.isas.ac.jp/suzaku/analysis/hxd/ gso-arf/

[7] ftp://legacy.gsfc.nasa.gov/docs/suzaku/analysis/ gso-bgd64bins.dat

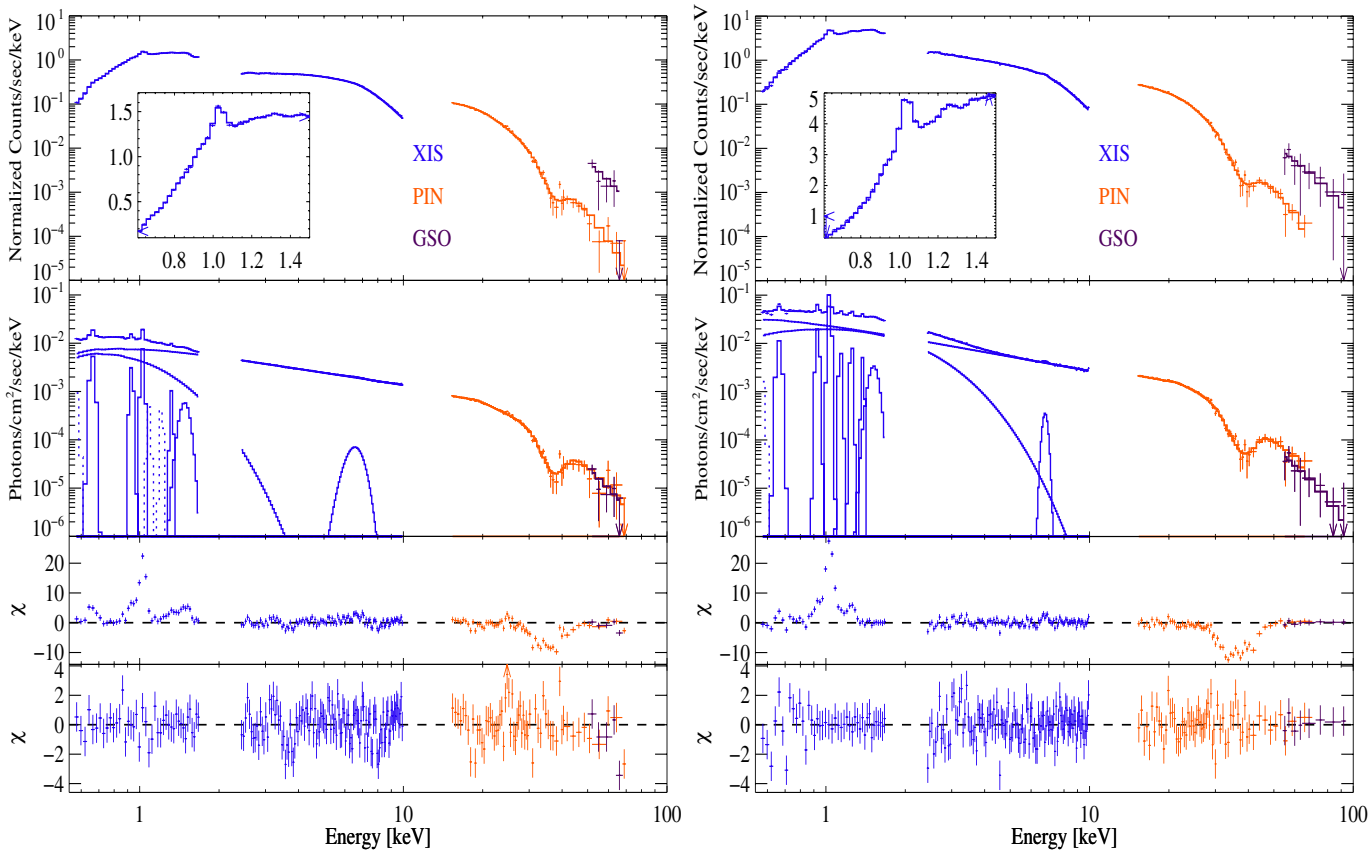


Fig. 1. *Top panels*: *Suzaku* broad-band spectrum before (*left panel*) and after (*right panel*) the 2008 torque reversal. Blue points denote XIS data (0.5−10 keV), PIN data points are in red (14−70 keV) and GSO in purple (50−70 keV and 50−100 keV, respectively). The residuals after fitting only the continuum model are shown in *the middle panel*, with no line emission and cyclotron line feature included. *The bottom panel* shows the residuals after fitting the line complex at ∼1 keV, the ∼6.5 keV iron line and the ∼37 keV cyclotron line (see Table 1 for a list of fitted parameters). A zoom of the region of the line complex at ∼1 keV is also shown for clarity.

Table 1. Best broad band fit spectral parameters[a] .

Observation	N_H[b]	α[c] norm	E_{cut}[d]	E_{Fold}[d]	E_{CRF}[d*] σ_{CRF}[d]	T_{BBody}[d] norm$_{BB}$[e]	Flux[f]	Constant factor[g]	χ^2_r(d.o.f.)
400015010 (Mar 2006)	1.46(2)	0.846(9) 0.010(1)	18.9(5)	13(1)	36.9 $^{+0.9}_{-0.8}$ 3.6(6)	0.229 $^{+0.01}_{-0.009}$ 127±13	0.55(9)	0.85(2) 0.8(fixed)	1.38(205)
405044010 (Sep 2010)	0.6(2)	0.981(3) 0.026(1)	21.6(7)	15(2)	37.8(8) 4.3(6)	0.511(8) 412 $^{+146}_{-108}$	1.58(4)	1.21(4) 0.8(5)	1.20(206)

Notes. [(a)] (PHABS (8 GAUSSIANS+GAUSSIAN+ BBODYRAD) +POW)*HIGHECUT *GABS)*CONST. [(b)] $\times 10^{21}$ cm^{-2}. [(c)] Photon Index with normalization units in photons keV^{-1} cm^{-2} s^{-1} at 1 keV. [(d)] keV. [(d*)] The values for the cyclotron line depths are 14 ± 4 and 18 ± 4 for the Mar. 2006 and Sep. 2010 observations, respectively. [(e)] With norm$_{BB}$ = $R^2_{BB_{km}}/d^2_{10}$, where d_{10} is the distance in units of 10 kpc, varying from 0.5−1.3 (assuming the 5−13 kpc distance range reported for this source). [(f)] $\times 10^{-9}$ erg cm^{-2} s^{-1} (0.5−100 keV), based on the XIS 03 flux calibration; XIS 03 is a combined spectrum from XIS 0 and XIS 3. [(g)] Intercalibration factors (PIN/XIS 03 and GSO/XIS 03). The GSO/XIS 03 factor was fixed to the value obtained for the second observation since it could not be well constrained.

(PHABS (BBODYRAD+POWLAW) HIGHECUT)*CONST. The constant factor (CONST) is used to account for intercalibration between XIS, PIN, and GSO. The Solar abundance table used in the plasma emission and photoelectric absorption models was set to wilm (Wilms et al. 2000). Spectral fits for both observations are shown in Fig. 1. A broad iron line near 6.4 keV (GAUSS) and a ∼37 keV cyclotron feature (GABS) improved the fit for both observations (see also Iwakiri et al. 2012). Table 1 presents the fitted values for these broad-band parameters from the final model (which included the low-energy lines described below).

Initial fitting results for the broad-band model for both observations indicated evidence for a strong line complex at ∼1 keV, shown in Fig. 1, middle panels. Following Angelini et al. (1995), we added seven narrow Gaussian lines (listed in Table 2) to our model. The line widths were fixed at the ASCA values (10 eV for all lines except the 1.362 keV line width was fixed at 57 eV). Table 2 lists the final fitted line parameters for both observations. Only five of the seven lines detected with ASCA were detected in the 2006 *Suzaku* observation, while six ASCA lines were detected in the 2010 *Suzaku* observation. For the tree ASCA lines

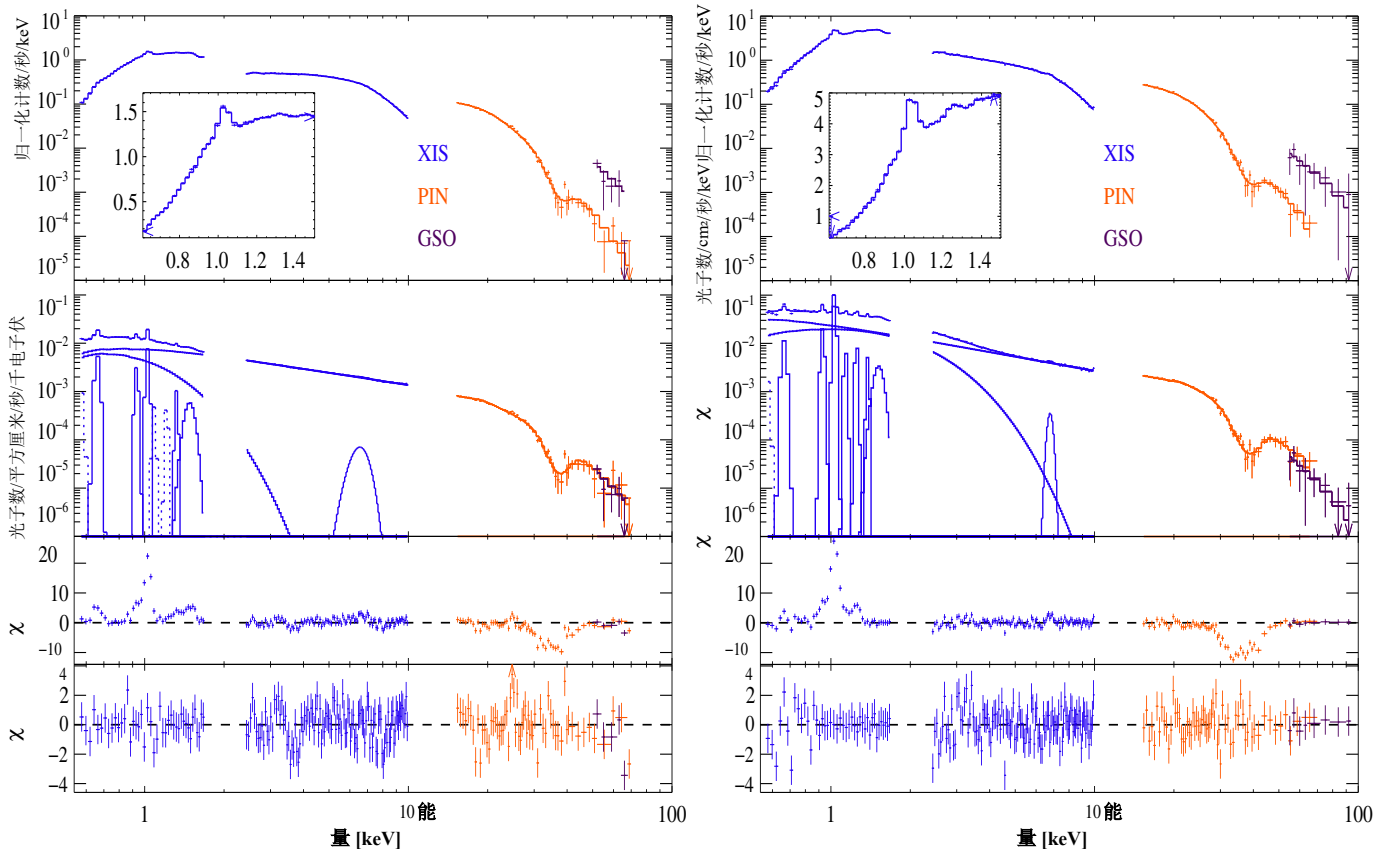


图 1. *顶部面板*：*Suzaku*在 2008 年扭矩反转前（*左面板*）和之后（*右面板*）的宽波段光谱。蓝色点表示 XIS 数据（0.5_10 keV），红色点为 PIN 数据点（14_70 keV），紫色点为 GSO 数据点（分别为 50_70 keV 和 50_100 keV）。仅拟合连续谱模型后的残差显示在*中间面板*中，未包含线发射和回旋线特征。*底部面板*显示了在拟合 ~1 keV 处的线复合体、~6.5 keV 铁线以及 ~37 keV 回旋线后的残差（拟合参数列表见表 1）。为清晰起见，图中还展示了 ~1 keV 处线复合体区域的放大图。

表 1.最佳宽带拟合光谱参数[a]。

观测	N_H[b]	α[c]	E_{cut}[d]	E_{Fold}[d]	E_{CRF}[d*]	T_{BBody}[d]	通量[f]	常数	χ^2_r（自由度）
	范数				σ_{CRF}[d]	$norm_{BB}$[e]		因子[g]		
400015010 （2006年3月）	1.46(2)	0.846(9) 0.010(1)	18.9(5)	13(1)	$36.9^{+0.9}_{-0.8}$ 3.6(6)	$0.229^{+0.01}_{-0.009}$ 127±13	0.55(9)	0.85(2) 0.8（固定值）	1.38(205)
405044010 (Sep 2010)	0.6(2)	0.981(3) 0.026(1)	21.6(7)	15(2)	37.8(8) 4.3(6)	0.511(8) 412^{+146}_{-108}	1.58(4)	1.21(4) 0.8(5)	1.20(206)

注。[a](PHABS (8 高斯函数+高斯函数+ bbodyrad) +POW)*highecut *GABS)*const.[b]×10^{21} cm^{-2}。[c]Photon Index，归一化单位为光子 keV^{-1} cm^{-2} s^{-1}，在 1 keV 处。[d]keV。[d*]回旋线深度的值对于 2006 年 3 月和 2010 年 9 月的观测分别为 14 ± 4 and 18 ± 4。[e]With $norm_{BB} = R^2_{BB_{km}}/d210$，其中$d_{10}$是以 10 kpc 为单位的距离，取值范围为 0.5_1.3（假设该源报告的 5_13 kpc 距离范围）。[f]×10^{-9} erg cm^{-2} s^{-1}(0.5_100 keV)，基于 XIS 03 的通量校准；XIS 03 是 XIS 0 和 XIS 3 的联合光谱。[g]交叉校准因子（PIN/XIS 03 和 GSO/XIS 03）。GSO/XIS 03 因子固定为第二次观测时获得的值，因为该值无法得到良好约束。

(PHABS (bbodyrad+ POWLAW) highecut)*const。常数因子（const）用于考虑XIS、PIN和GSO之间的相互校准。用于等离子体发射和光电吸收模型的太阳丰度表设定为wilm（Wilms等人，2000年）。两次观测的光谱拟合结果如图1所示。位于6.4 keV附近的宽铁线（高斯分布）和约37 keV的回旋特征（GABS）改善了两次观测的拟合效果（另见Iwakiri等人2012）。表1列出了最终模型中这些宽带参数的拟合值（该模型还包括下文所述的低能线）。

两次观测的宽带模型初步拟合结果表明，在~1 keV处存在一个强谱线复合体的证据，如图1中间面板所示。遵循Angelini等人（1995）的方法，我们在模型中添加了七条窄高斯线（列于表2中）。谱线宽度固定为 ASCA 值（除1.362 keV谱线外，其余所有谱线宽度固定为10 eV；1.362 keV谱线宽度固定为57 eV）。表2列出了两次观测的最终拟合谱线参数。在通过 ASCA 检测到的七条谱线中，仅有五条在2006年*Suzaku*观测中被检测到；而在2010年*Suzaku*观测中则检测到六条 ASCA 谱线。对于这六条 ASCA 谱线

Table 2. Fit results on line emission.

Observation	Expected (keV)	Line energy (keV) Mar. 2006	Sep. 2010	$\Delta\chi^2$ Mar. 2006 (d.o.f. = 205)	Sep. 2010 (d.o.f. = 206)	Intensity[a] Mar. 2006	Sep. 2010	*EW* (eV) Mar. 2006	Sep. 2010
O Heα	0.568	0.568(fixed)	0.568(fixed)	–	–	0.09(fixed)	0.09(fixed)	3.2(2)	1.3(2)
O Lyα	0.653	0.661(8)	0.66(2)	59	5.4[b]	$3.0^{+1}_{-0.9}$	4(4)	13(4)	7(3)
Ne Heα[c]	0.914	0.930(8)	0.929(8)	105	119	1.3(3)	7(1)	8(1)	14(1)
Ne Lyα	1.021	1.025(3)	1.031(2)	878	1699	3.3(2)	24(1)	22.6(2)	50.9(8)
Ne Heβ[c]	1.084	1.084(fixed)	1.13(3)	–	39	0.2(fixed)	3(1)	3.7(2)	6(2)
Ne Lyβ	1.210	1.210(fixed)	1.25(1)	6	96	0.2(fixed)	2(1)	1.8(2)	9(4)
Mg Heα	1.34	1.32(2)	1.38(2)	36	31	0.4(1)	$1.8^{+0.9}_{-1}$	4(1)	5(1)
Ne Recombo[d]	1.362	1.48(2)	1.52(2)	137	154	0.9(2)	5(1)	11.1(3)	15(5)
Fe K-fluor. line[e]	6.4	6.5(2)	6.78(7)	59	30	$0.8^{+0.4}_{-0.3}$	1.3(5)	42.2(3)	33(1)

Notes. The widths of the low-energy lines were fixed to the ASCA values (10 eV for all but the 1.362 keV line with 57 eV). [(a)] photons 10^{-4} cm^{-2} s^{-1}. [(b)] We note that, for the 2010 observation, the $\Delta\chi^2$ for this line is not well computed due to the spectral model not being able to adequately fit this line (see Fig. 2, right panel). [(c)] The abundances of the lines at 0.914 and 1.084 keV can also be consistent with a blend of several *L*-shell iron line emission (see Angelini et al. 1995). [(d)] The Mg Hα line emission occurs at 1.471 keV (here not resolved, and thus not detected, by *Suzaku*), and therefore this Mg line may be contributing to this feature. [(e)] For the iron line the widths were fitted (0.5(3) keV and 0.14(7) keV, respectively).

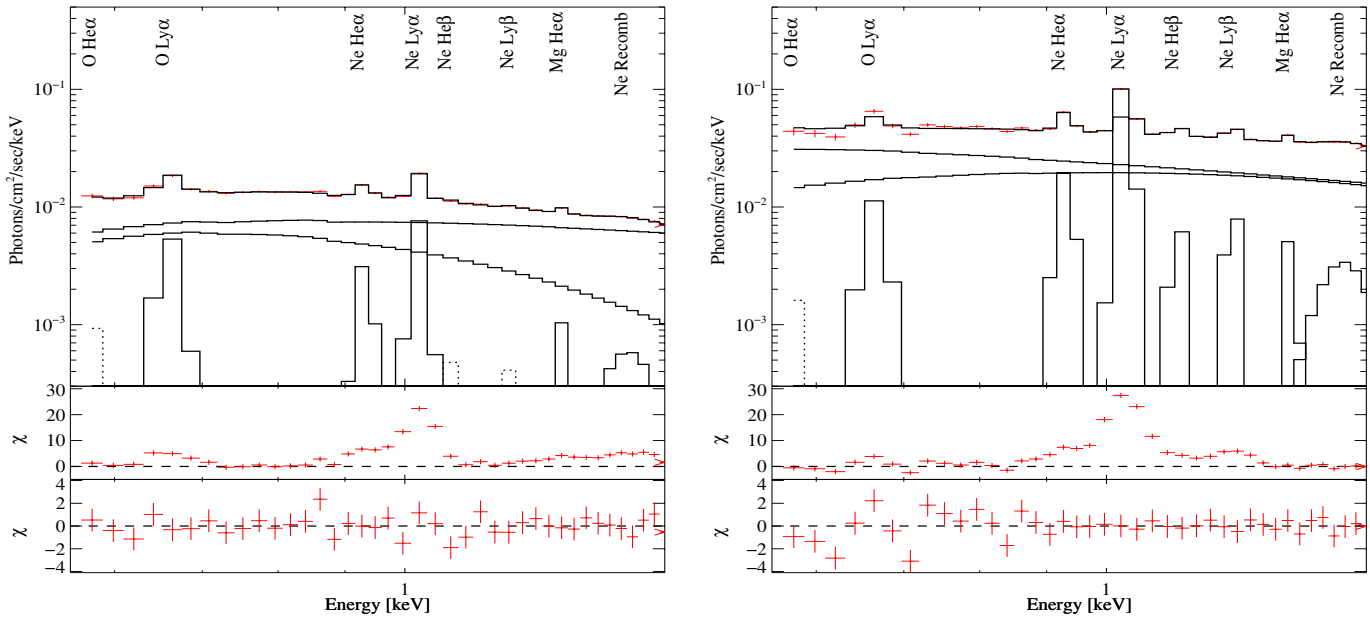


Fig. 2. *Top panels*: Eight Gaussian lines from the ∼1 keV complex detected by *Suzaku*, before (*left panel*) and after (*right panel*) the 2008 torque reversal. Here the dotted lines in the model denote the components that were fixed. The residuals after fitting only the continuum model are shown in *the middle panel*, with no line emission included. *The bottom panel* shows the residuals after fitting the line complex with those 8 lines (see Table 2 for more information about these lines).

not well detected in 2006 and the one in 2010 the intensities were fixed, since the errors could not be well constrained.

The broad feature at 1.362 keV could alternatively be explained by a recombination edge. Fits to both observations using a model that replaced the 1.362 keV Gaussian line with a recombination edge yielded similar $\delta\chi^2$ to our model containing the 1.362 keV line. The edge (and temperature) obtained for the 2006 and 2010 observations are 1.45 ± 0.05 keV (42^{+50}_{-40} eV) and 1.47 ± 0.03 keV (45^{+34}_{-25} eV) (fitting with a Gaussian line gives 1.48(2) and 1.51(2) keV, see Table 2), respectively. A posteriori inspection of the residuals for both observations, suggested the presence of an extra Gaussian component at energy ∼1.34 keV. Therefore, up to 8 Gaussian components were included in the fit (see Fig. 2; $\Delta\chi^2$ of 283 for 205 d.o.f., and 248 for 206 d.o.f., for the first and second observations, respectively). Final fit parameters for this line at ∼1.34 keV, tentatively identified as the He-like Mg, are listed in Table 2. The $\Delta\chi^2$ obtained by setting the flux of a particular line to zero in the best-fit model indicates the significance of each line. In addition, we fit the broad-band spectrum using different combinations of the XIS detectors yielding similar results. Other models accounting for emission from a hot diffuse gas, with line emissions coming from several elements, were also fit (*meka* and *vmeka* in XSPEC notation). We changed the elemental abundances while performing the fits with the *meka* and *vmeka* models, however they did not give reasonable fits.

A luminosity increase occurred between the two observations, changing from $L_{(0.5-10\ \mathrm{keV})} \approx 2.4 \times 10^{34}\,\mathrm{d}^2_{\mathrm{kpc}}$ to $6 \times 10^{34}\,\mathrm{d}^2_{\mathrm{kpc}}$ erg s^{-1}. However, the value of the center of energy of the cyclotron line did not exhibit a significant change. This

表 2.线排放的拟合结果。

	预期线能（keV） 观测值（keV）：2006年3月、2010年9月；2006年3月、2010年9月；2006年3月、2010年9月；2006年3月、2010年9月 (d.o.f. = 205) (d.o.f. = 206)			$\Delta\chi^2$		强度[a]		*EW* (eV)	
He**α**	0.568	0.568（固定值）	0.568（固定值）	–	–	0.09（固定值）	0.09（固定值）	3.2(2)	1.3(2)
O Lyα	0.653	0.661(8)	0.66(2)	59	5.4[b]	$3.0^{+1}_{-0.9}$	4(4)	13(4)	7(3)
Ne Heα[c]	0.914	0.930(8)	0.929(8)	105	119	1.3(3)	7(1)	8(1)	14(1)
Ne Lyα	1.021	1.025(3)	1.031(2)	878	1699	3.3(2)	24(1)	22.6(2)	50.9(8)
Ne Heβ[c]	1.084	1.084（固定）	1.13(3)	–	39	0.2（固定值）	3(1)	3.7(2)	6(2)
Ne Lyβ	1.210	1.210（固定）	1.25(1)	6	96	0.2（固定值）	2(1)	1.8(2)	9(4)
Mg Heα	1.34	1.32(2)	1.38(2)	36	31	0.4(1)	$1.8^{+0.9}_{-1}$	4(1)	5(1)
Ne Recombo[d]	1.362	1.48(2)	1.52(2)	137	154	0.9(2)	5(1)	11.1(3)	15(5)
Fe K-荧光线[e]	6.4	6.5(2)	6.78(7)	59	30	$0.8^{+0.4}_{-0.3}$	1.3(5)	42.2(3)	33(1)

注。低能谱线的宽度被固定为 ASCA 值（除 1.362 keV 谱线为 57 eV 外，其余均为 10 eV）。[(a)]光子通量为 10^{-4} cm^{-2} s^{-1}。[(b)]我们注意到，对于 2010 年的观测，由于光谱模型无法充分拟合该谱线，因此该谱线的 $\Delta\chi^2$计算结果不够准确（见图2右图）。[(c)]0.914 和 1.084 keV 处谱线的丰度也可能与多种*L*层铁谱线发射的混合模型相吻合（参见Angelini 等人1995）。[(d)]Mg H**α** 谱线发射发生在 1.471 keV 处（此处未被*Suzaku*分辨，因此未被探测到），因此该 Mg 谱线可能对该特征有所贡献。[(e)]对于铁谱线，其宽度经过拟合（分别为 0.5(3) keV 和 0.14(7) keV）。

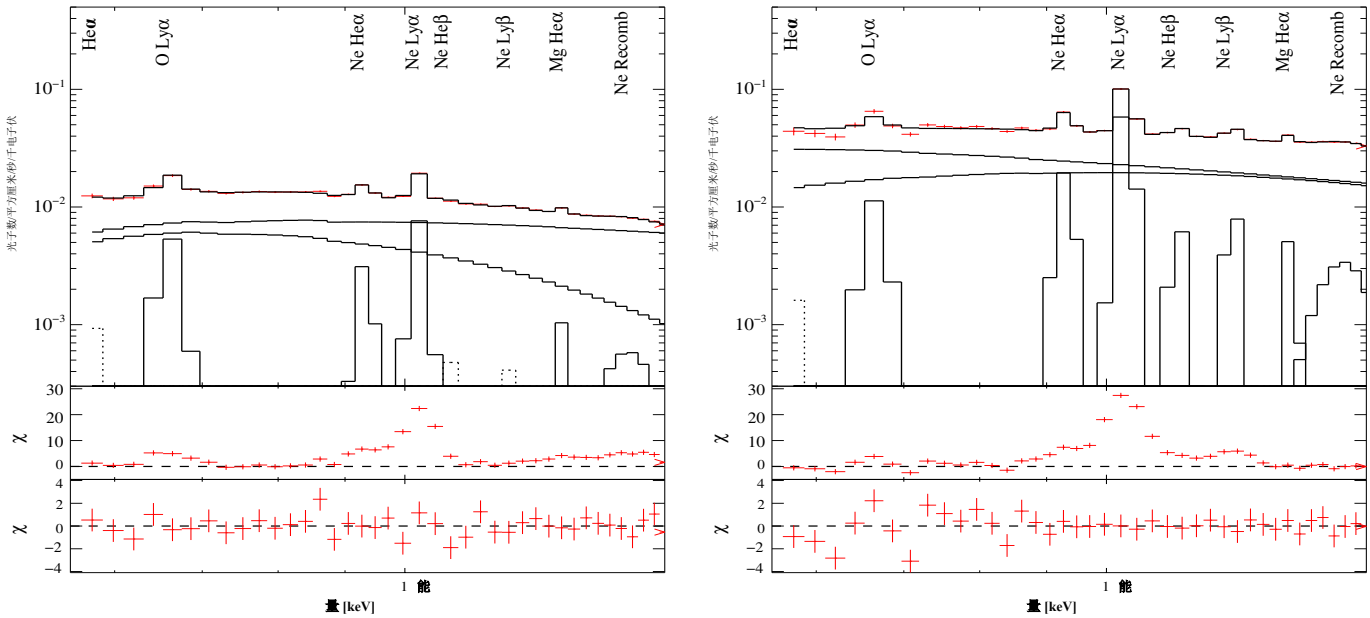


图 2。*顶部面板*：由*Suzaku*探测到的来自 ~1 keV 复合线的八条高斯线，在 2008 年扭矩反转之前（*左面板*）和之后（*右面板*）。此处模型中的虚线表示被固定的分量。仅拟合连续谱模型后的残差显示在*中间面板*中，未包含任何线发射。*底部面板*显示了使用这 8 条线拟合线复合体后的残差（有关这些线的更多信息，请参阅表2）。

2006 年的结果未能被有效检测出来；而 2010 年的结果中，强度值被固定为固定值，因为误差无法得到充分约束。

1.362 keV处的宽特征也可以由复合边来解释。使用将 1.362 keV高斯线替换为复合边的模型对两次观测数据进行拟合，得到的$\delta\chi^2$与包含1.362 keV谱线的模型结果相近。针对2006年和2010年观测数据得到的能级边（及对应温度）分别为 1.45 ± 0.05 keV（42 $-^{+50}_{40}$ eV）和 1.47 ± 0.03 keV（45 $-^{+34}_{25}$ eV）（若采用高斯线拟合则分别得到1.48(2)和1.51(2) keV，见表2）。对两次观测数据的残差进行事后检验，表明在能量约1.34 keV处存在一个额外的高斯分量。因此，拟合过程中最多纳入了8个高斯分量（见图2；对于第一次和第二次观测，$\Delta\chi^2$分别为283（自由度为205）和248（自由度为206））。该位于约1.34 keV处的谱线的最终拟合参数（暂被识别为类He的Mg）列于表2中。通过将特定谱线的通量设为零并在最佳拟合模型中获得的$\Delta\chi^2$，可反映各谱线的显著性。此外，我们使用不同的XIS探测器组合对宽带光谱进行了拟合，得到了相似的结果。其他考虑热弥散气体辐射、且谱线辐射来自多种元素的模型也进行了拟合（*meka*和*vmeka*，以 XSPEC 符号表示）。我们在使用*meka*和*vmeka*模型进行拟合时改变了元素丰度，但结果并未得到合理的拟合效果。

在两次观测之间，光度发生了增加，从$L_{(0.5\text{-}10\ \mathrm{keV})} \approx 2.4 \times 10^{34}\, d_{\mathrm{kpc}}^2$变为 $6 \times 10^{34}\, d_{\mathrm{kpc}}^2$ erg s^{-1}。然而，回旋线能量中心的值并未表现出显著变化。这

behavior resembles the case of the Be/X-ray binary system A 0535+26 (Caballero et al. 2009), but is contrary to sources like Her X–1 and V 0332+53 (Pottschmidt et al. 2012, and references therein) or GX 304–1 (Klochkov et al. 2012), which show positive or negative L–$E_{\rm CRF}$ correlations (see also Becker et al. 2012). On the other hand, the column density, $N_{\rm H}$, dropped by a factor of ∼3 while the black body temperature increased by a factor of ∼2. The rest of the broadband parameters were consistent between the two observations. Absorbed fluxes in the 0.5–10 keV, 2–10 keV, 2–20 keV and 2–50 keV bands are 0.18(4), 0.159(2), 0.295(3), 0.51(3) × 10^{-9} erg cm^{-2} s^{-1} and 0.49(3), 0.38(9), 0.64(1), 1.3(9) × 10^{-9} erg cm^{-2} s^{-1}, for the 2006 and 2010 *Suzaku* observations, respectively.

4. Discussion and conclusions

4.1. Spectral continuum

Our results show that two years after the torque reversal in 2008 the luminosity has reached almost the same level as in 1977 (see Fig. 3), the black body temperature has increased to ∼0.6 keV and the pulsar shows the same spin-up rate. The system's appearance in this state can be explained in terms of accretion onto the neutron star from a geometrically thin Keplerian disk provided the mass-transfer rate is limited to $\dot{M} \geq 10^{16}$ g s^{-1} (Chakrabarty et al. 1997).

The pulsar's transition from spin-up to spin-down in 1990 was accompanied by significant changes of spectral parameters. The black body temperature decreased from ∼0.6 keV to ∼0.3 keV and the power law photon index changed from ∼1.5 (Vaughan & Kitamoto 1997) to ∼0.6 (Vaughan & Kitamoto 1997; Angelini et al. 1995; Owens et al. 1997). The transition period lasted about 150 days and the source luminosity during this time was only a factor of 2 smaller than its highest level observed in 1977 (see Fig. 3). This indicates that the mass-transfer rate during the torque reversal in 1990 was in excess of 5×10^{15} g s^{-1}.

Our *Suzaku* observations show that the pulsar's behavior during the torque reversal in 2008 (from spin-down to spin-up phase) was different. While the transition period also lasted about 150 days and the black body temperature changed from ∼0.23 keV back to ∼0.6 keV in 2010, the photon index remained almost unchanged. The value of the order of unity observed in 2006 and 2010 is the same as the photon index observed by several missions before the torque reversal during the spin-down phase, e.g. ASCA (Angelini et al. 1995), *BeppoSAX* (Orlandini et al. 1998), *Chandra* and XMM (Krauss et al. 2007; Schulz et al. 2001). The torque reversal in 2008 occurred at slightly lower luminosity level than in 1990 and was followed by a brightening of the X-ray source on a time scale of more than 2 years.

As we have shown in a previous paper (Camero-Arranz et al. 2010), the torque reversals in 4U 1626–67 are unlikely to be associated with alternation of the accretion flow geometry between the Keplerian geometrically thin disk (during the spin-up phase) and a spherical flow or a hot, geometrically thick disk (during the spin-down phase). The corresponding scenarios (see, e.g. Yi & Vishniac 1999) encounter major difficulties explaining 150 day transition period (see also Wijers & Pringle 1999) and the low level of X-ray flux measured before and after the torque reversal in 2008. Furthermore, the mass-transfer rate during both of the observed torque reversals ($\geq 5 \times 10^{15}$ g s^{-1}) was above the critical mass-transfer rate at which the transition of the accretion disk to the hot, geometrically thick state can be expected (Narayan & Yi 1995). Finally, observations of the strong emission-line complex

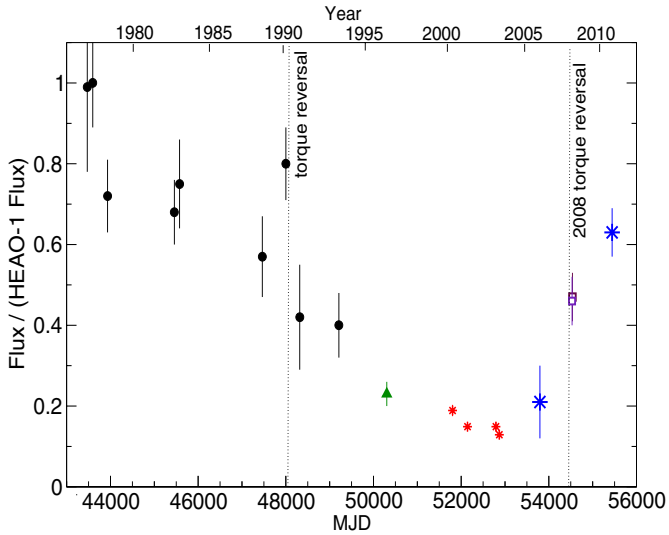


Fig. 3. The X-ray flux history of 4U 1626–67 relative to the flux measured by HEAO 1, in the same energy band, from previous works (Chakrabarty et al. 1997: circles; Orlandini et al. 1998: triangle; Krauss et al. 2007: stars; Camero-Arranz et al. 2011, unfilled squares) and the two *Suzaku* observations (blue diamonds) in the 0.7–60 keV band.

centered on 1 keV and the iron K-fluorescence line at 6.4 keV during both spin-up and spin-down phases indicate that the temperature of the material surrounding the neutron star does not exceed a few keV. Conversely, the temperature of the material in a thick disk is close to the adiabatic temperature, which for the parameters of 4U 1626–67 is in excess of 10 keV. Thus, our results favor a scenario in which the neutron star in 4U 1626–67 accretes material from a geometrically thin disk during both the spin-up and spin-down phases.

The magnetospheric radius of a neutron star accreting material from a geometrically thin disk can be evaluated as $r_{\rm m} \simeq \kappa\left(\mu^2/\dot{M}(2GM_{\rm ns})^{1/2}\right)^{2/7}$, where $\mu = (1/2)B_* R_{\rm ns}^3$ is the dipole magnetic moment, B_* the surface field and $R_{\rm ns}$ the radius of the neutron star. κ is a parameter accounting for the geometry of the accretion flow, which in the case of disk accretion is limited to $\kappa \geq\sim\ 0.5$ (Ghosh & Lamb 1979; Wang 1996; Bozzo et al. 2009). For the star to be in the accretor state its magnetospheric radius should not exceed the corotation radius, $r_{\rm cor} = \left(GM_{\rm ns}P_{\rm s}^2/4\pi^2\right)^{1/3}$. Otherwise, the centrifugal barrier at the magnetospheric boundary would prevent the accretion flow from reaching the stellar surface (Shvartsman 1970). This condition in the case of 4U 1626–67 is satisfied if the mass-transfer rate is $\dot{M} \geq \dot{M}_{\rm min}$, where

$$\dot{M}_{\rm min} \simeq 6 \times 10^{14}\ {\rm g\ s^{-1}}\ \kappa_{0.5}^{7/2} m^{-5/3} P_{7.7}^{-7/3} \mu_{30}^2. \quad (1)$$

Here $\kappa_{0.5} = \kappa/0.5$ is normalized to its minimum value, m is the mass and $P_{7.7}$ the spin period of the neutron star in units of 1.4 $M_\odot$ and 7.7 s. The dipole magnetic moment of the neutron star, μ_{30}, is normalized to 10^{30} G cm^3. According to our measurements of the energy of the cyclotron line feature with *Suzaku* ($36.8^{+0.9}_{-0.8}$ keV) the surface field of the neutron star in 4U 1626−67 is ∼3 × 10^{12} G. This is consistent with previous estimate of the field strength made by Orlandini et al. (1998) from observations of 4U 1626–67 with *BeppoSAX*. Putting this value to Eq. (1) and taking into account that 4U 1626–67 is a persistent pulsar one finds that the mass-transfer rate has never decreased below 10^{15} g s^{-1} in the time period between 1977

其行为类似于Be/X射线双星系统A 0535+26（Caballero等人2009），但与Her X−1和V 0332+53（Pottschmidt等人2012及其参考文献）或 GX 304−1（Klochkov等人2012）等源相反，这些源显示出正或负的L_E_{CRF}相关性（另见Becker等人2012）。另一方面，柱密度N_H下降了约3倍，而黑体温度则上升了约2倍。其余宽带参数在两次观测中保持一致。对于2006年和2010年*Suzaku*观测，0.5_10 keV、2_10 keV、2_20 keV和2_50 keV能段的吸收流量分别为0.18(4)、0.159(2)、0.295(3)、0.51(3) × 10^{-9} erg cm^{-2} s^{-1}，以及0.49(3)、0.38(9)、0.64(1)、1.3(9) × 10^{-9} erg cm^{-2} s^{-1}。

4. 讨论与结论

4.1. 光谱连续谱

我们的结果表明，2008年扭矩反转两年后，光度已接近1977年的水平（见图3）；黑体温度已升至约0.6 keV；脉冲星的自转加速率也保持不变。该系统处于此状态的原因，可以用几何薄开普勒盘向中子星吸积物质来解释，前提是质量转移率被限制在$\dot{M} \geq 10^{16}$ gs^{-1}以内（Chakrabarty等人，1997）。

该脉冲星在1990年从自转加速阶段转变为自转减速阶段时，其光谱参数也发生了显著变化。黑体温度从约0.6 keV降至约0.3 keV，幂律光子指数则从约1.5（Vaughan & Kitamoto 1997）变为约0.6（Vaughan & Kitamoto 1997；Angelini et al.1995；Owens et al.1998）。这一转变期持续了约150天，期间源的光度仅比1977年观测到的峰值低约2倍（见图3）。这表明1990年力矩反转期间的物质转移率超过了5×10^{15} gs^{-1}。

我们的*Suzaku*观测表明，该脉冲星在2008年扭矩反转期间（从自转减速阶段到自转加速阶段）的行为有所不同。虽然过渡期也持续了约150天，且黑体温度在2010年从约0.23 keV恢复到约0.6 keV，但光子指数几乎保持不变。2006年和2010年观测到的数量级为1的值，与在自转减速阶段扭矩反转前由多个任务观测到的光子指数相同，例如ASCA（Angelini 等人 1995）、*BeppoSAX*（Orlandini 等人 1998）、*Chandra*和 XMM（Krauss等人2007；Schulz等人2001）。2008年的扭矩反转发生在比1990年略低的亮度水平，随后X射线源在超过2年的时间尺度上出现增亮现象。

正如我们在先前的一篇论文中所示（Camero-Arranz 等人2010），4U 1626−67 中的扭矩反转不太可能与吸积流几何形态在开普勒几何薄盘（自转加速阶段）与球形流或高温几何厚盘（自转减速阶段）之间的交替有关。相应的情景（参见例如Yi & Vishniac 1999）在解释 150 天过渡期（另见Wijers & Pringle 1999）以及 2008 年扭矩反转前后测得的低水平 X 射线通量方面遇到了重大困难。此外，在两次观测到的扭矩反转期间，质量转移率 (≥5 × 10^{15} gs^{-1}) 均高于临界质量转移率，而在此临界值下，可以预期吸积盘会向高温几何厚态转变（Narayan & Yi 1995）。最后，对以 1 keV 为中心的强发射线复合体以及位于 6.在自转加速阶段和自转减速阶段均为 4 keV，表明中子星周围物质的温度未超过几 keV；相反，厚盘内物质的温度接近绝热温度，对于 4U 1626−67 的参数而言，该温度超过 10 keV。因此，我们的结果支持这样一种情景：4U 1626−67 中的中子星在自转加速阶段和自转减速阶段均从几何薄盘中吸积物质。

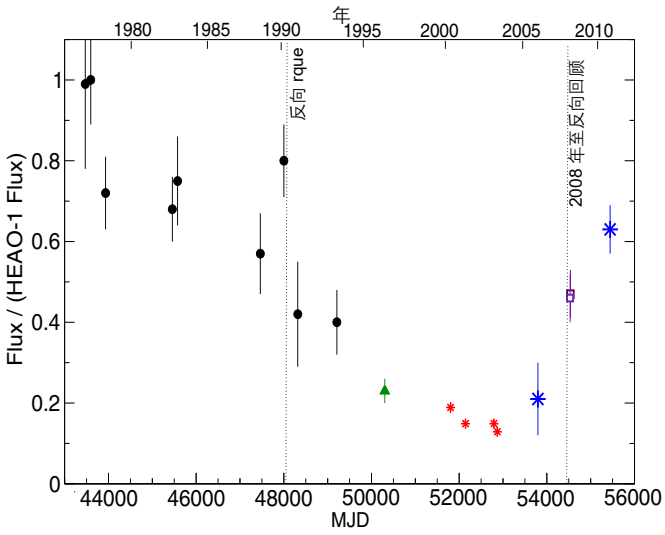


图 3.4U 1626−67 的 X 射线流量历史，相对于 HEAO 1 在相同能段内测得的流量（来自先前研究：Chakrabarty 等人1997：圆圈；Orlandini 等人1995：三角形；Krauss 等人2007：星形；Camero-Arranz 等人 2011，未填充方块）以及两次*Suzaku*观测（蓝色菱形）在 0.7_60 keV 能段内的数据。

从几何薄吸积盘吸积物质的中子星磁层半径可评估为r_m~κ(μ $^2/\dot{M}$. $(2GM_{ns})^{1/2})^{2/7}$，其中 μ = $(1/2)B_*R_{ns}^3$为偶极磁矩，B_*为表面磁场，R_{ns}为中子星半径。κ 是用于描述吸积流几何形状的参数，在盘状吸积情形下其取值上限为 κ ≥~ 0.5（Ghosh & Lamb 1979；Wang 1996；Bozzo 等人2009）。为使中子星处于吸积态，其磁层半径不应超过共转半径，$r_{cor} = \left(GM_{ns}P_s^2/4\pi^2\right)^{1/3}$。否则，磁层边界处的离心势垒会阻止吸积流到达恒星表面（Shvartsman 1970）。对于 4U 1626−67，若质量转移率满足$\dot{M} \geq \dot{M}_{min}$，则上述条件成立，其中

$$\dot{M}_{min} \simeq 6 \times 10^{14}\ \mathrm{g\ s^{-1}}\ \kappa_{0.5}^{7/2} m^{-5/3} P_{7.7}^{-7/3} \mu_{30}^2. \quad (1)$$

此处$\kappa_{0.5}$=κ/0.5 被归一化至其最小值，m为质量，而$P_{7.7}$为中子星的自转周期，单位为1.4 $M_\odot$和7.7秒。中子星的偶极磁矩μ_{30}被归一化至10 ^{30} G·cm ^{3}。根据我们利用*Suzaku*对回旋线特征能量（36.8 $^{+0.9}_{-0.8}$keV）的测量结果，4U 1626_67 中子星的表面磁场强度约为3 × 10^{12} G。这与Orlandini 等人（1998）通过*BeppoSAX*对4U 1626−67的观测所作出的磁场强度估算结果一致。将该值代入等式（1），并考虑到4U 1626−67是一个持续脉冲星，可以得出质量转移率在1977年以来的时间段内从未降至10 ^{15} g·s^{-1}以下。

and 2010. The rate of mass-transfer, therefore, may vary in the system up to an order of magnitude. It is remarkable that the observed X-ray flux (and, correspondingly, the mass accretion rate onto the surface of the neutron star) varies with similar amplitude (see Fig. 3). This indicates that either the value of κ is indeed close to its lower limit, or the flux variations are governed by a different reason. In particular, one can also envisage a situation in which the magnetospheric radius of the neutron star during the low luminosity phase is close to the corotation radius (this implies κ to be substantially larger than its minimum possible value) and the centrifugal barrier at the magnetospheric boundary reduces the mass transfer rate into the stellar magnetosphere (for discussion see Perna et al. 2006). The mass-transfer rate in the disk within this scenario varies insignificantly as the pulsar switches between the spin-up and spin-down phases.

While the black body temperature during 2006–2010 increased from 0.23 keV to 0.51 keV, the source luminosity increased only by a factor of ∼2. This implies that during the brightening phase the effective area of the black-body source has decreased by almost an order of magnitude. A high luminosity and a relatively small size of this source suggest that it is located inside the magnetosphere in the vicinity of the neutron star. The accretion process in this region is fully controlled by the stellar magnetic field and plasma flows within the accretion channel. In this light, variations of the effective area of the black-body source can be associated with a restructuring of the accretion channel (i.e. change of both its cross-section and the height of the shock at the base of the accretion column). Studies (see e.g. Burnard et al. 1983; Basko & Sunyaev 1976; Kulkarni & Romanova 2008) suggest that the parameters of the channel depend on the geometry and physical conditions in the accretion flow at the magnetospheric boundary, as well as on the mode by which the material enters the magnetic field of the neutron star. Our observations give no evidence of any significant change in the accretion flow geometry during the brightening phase (see above). It, therefore, appears more plausible to associate the restructuring of the accretion channel with change of the mode by which the accreting material enters the magnetosphere.

It is widely adopted that the accreting material can enter the magnetosphere due to the Rayleigh-Taylor and Kelvin-Helmholtz instabilities or/and the magnetic reconnection. The Rayleigh-Taylor instability can be at work if the angular velocity of the accreting material at the boundary does not significantly differ from the angular velocity of the neutron star itself. The magnetic field configuration in the magnetopause in this case is close to poloidal. If this condition is not satisfied the Rayleigh-Taylor instability can be suppressed by the magnetic field shear in the magnetopause (Burnard et al. 1983; Ikhsanov & Pustil'nik 1996). This indicates that the relative velocity between the magnetic field of the neutron star and the accreting material at the boundary can be one of the key parameters determining the structure of the accretion channel. Even small fluctuations in the mass transfer rate may lead to significant variations of this parameter if the magnetospheric radius is close to the corotation radius (see e.g. Perna et al. 2006). Another possibility is an amplification of the magnetic field at the inner radius of the disk governed by the dynamo action (see e.g. Rappaport et al. 2004). The disk in this case may switch into a sub-Keplerian state and decrease the angular velocity of the material at its inner radius. However, the modeling of the evolution of the channel, and its influence on the properties of the black body source within these scenarios, is very complicated and beyond the scope of the present paper.

4.2. Emission lines

The observed change of the equivalent width (EW) of the Ne Lyβ line at 1.021 keV during the brightening phase of the pulsar from 22.6 to 50.9 eV, as well as the increase of its intensity a factor of ∼8 reflect a transition of the accretion picture to a new phase. This may indicate either changes of plasma parameters at the inner radius of the accretion disk (Kii et al. 1986) or changes in the wind caused by interaction between the Keplerian disk and the stellar magnetic field at the corotation radius (Lovelace et al. 1995). As shown by Perna et al. (2006), the mass outflow rate can be higher during the spin-down phase and lead to a periodic circulation of material between the inner and outer radii of the accretion disk. This feedback results in a redistribution of mass in the accretion disk and can lead to variations of the mass-transfer rate in the system. If this is the cause of the observed luminosity increase after the torque reversal in 2008, the time scale of radial motion of the material in the disk is $\tau_r \sim 2$ years. This situation can be realized if the α parameter at the outer radius of the disk is relatively small,

$$\alpha \sim 0.02\, m^{1/2} \left(\frac{c_s}{10^6 \text{ cm s}^{-1}}\right)^{-2} \left(\frac{\tau_r}{2 \text{ yr}}\right)^{-1} \left(\frac{R}{10^{10} \text{ cm}}\right)^{1/2}, \quad (2)$$

and, correspondingly, the density of material in the disk significantly exceeds its average value for X-ray pulsars (for discussion see e.g. King 2012). Here c_s is the sound speed in the disk at the distance R. The slow turbulization of the material in the disk can also explain the small value of the strength of the torque noise observed from 4U 1626–67 (Chakrabarty et al. 1997).

In the present work, following Angelini et al. (1995), we have decomposed the strong emission-line complex centered on 1 keV found in the *Suzaku* X-ray spectrum of 4U 1626–67 (see Fig. 2). In both *Suzaku* observations the strongest line is the hydrogen-like Ne Lyα line at ∼1.025 keV, in agreement with the ASCA observations (Angelini et al. 1995). The broad feature at ∼1.48 keV found in our observations is close to the expected energy of the H-like Ne recombination edge at 1.362 keV. The measured 57 eV broadening of the line at ∼1.48 keV suggests that we may be also detecting the recombination continuum of Ne, as suggested by Angelini et al. (1995) for the ASCA observation. We point out that the Mg Hα transition occurs at energy 1.471 keV (which has not been resolved by *Suzaku*), therefore this Mg line may be blended with the recombination continuum of Ne. Furthermore, with *Suzaku* an eighth line centered at ∼1.34 keV was detected, which we tentatively identify as the Mg Heα (see Table 2).

During the spin-down period, comparing the strength of the low-energy lines detected in the Mar. 2006 *Suzaku* observation with previous measurements by *ASCA*, we see that the overall decrease is of the order of ∼2.5 (see Table 2). Before the 2008 torque reversal, X-ray missions like *BeppoSAX* (Owens et al. 1997), *Chandra* (Schulz et al. 2001; Krauss et al. 2007) and XMM (Krauss et al. 2007) were able to resolve only up to 3 or 4 lines of this complex. In general, the measurements of the intensity of those lines are in agreement with our results. During the new spin-up period after the 2008 torque reversal, the flux of the lines increased in general by a factor of ∼5 except the Ne Lyα line at 1.021 keV that increased by a factor of ∼8. This result is not surprising taking into account such a change in luminosity. Unfortunately, during a previous spin-up period there was only one measurement of the ∼1 keV complex by the *Einstein* observatory in 1979, and it was not resolved due to the lower data quality. The overall luminosity decrease associated with the 1990 torque reversal was a factor of ∼3

以及 2010 年。因此，系统中的质量转移率可能相差一个数量级。值得注意的是，观测到的 X 射线通量（以及相应地，中子星表面的质量吸积率）也以类似的幅度发生变化（见图3）。这表明，要么 κ 的值确实接近其下限，要么通量的变化是由其他原因决定的。具体而言，还可以设想这样一种情况：在中子星低光度阶段，其磁层半径接近共转半径（这意味着 κ 显著大于其理论最小值），而磁层边界处的离心势垒会降低进入恒星磁层的质量转移率（相关讨论见Perna 等人2006）。在此情景下，当脉冲星在自旋加速阶段与自旋减速阶段之间切换时，盘内的质量转移率变化极小。

尽管在2006_2010年期间，黑体温度从0.23 keV上升至0.51 keV，但源光度仅增加了约2倍。这表明，在增亮阶段，黑体源的有效面积几乎缩小了一个数量级。该源的高光度及相对较小的尺寸表明，它位于中子星附近的磁层内部。此区域的吸积过程完全受恒星磁场及吸积通道内等离子体流的控制。由此看来，黑体源有效面积的变化可能与吸积通道的重构有关（即吸积柱底部的横截面积和激波高度均发生变化）。相关研究（例如Burnard等人1983；Basko与Sunyaev 1976；Kulkarni与Romanova 2008）指出，通道参数取决于磁层边界处吸积流的几何构型与物理条件，以及物质进入中子星磁场时所采用的模式。中子星。我们的观测结果并未发现，在增亮阶段（见上文）吸积流几何结构发生任何显著变化的证据。因此，将吸积通道的重构与吸积物质进入磁层的方式发生变化联系起来，似乎更为合理。

普遍认为，吸积物质可通过瑞利-泰勒不稳定性、开尔文-亥姆霍兹不稳定性或/和磁重联进入磁层。若吸积物质在边界处的角速度与中子星自身的角速度无显著差异，则瑞利-泰勒不稳定性可能发挥作用；此时磁层顶部的磁场构型接近极向型。若该条件不满足，瑞利-泰勒不稳定性可被磁层顶部的磁场剪切所抑制（Burnard 等人1983；Ikhsanov 与 Pustil ′nik 1996）。这表明，中子星磁场与边界处吸积物质之间的相对速度，可能是决定吸积通道结构的关键参数之一。若磁层半径接近共转半径，即使质量转移率存在微小波动，也可能导致该参数发生显著变化（参见例如 Perna 等人2006）。另一种可能性是，由发电机作用主导的盘内半径处的磁场被增强（参见例如Rappaport 等人2004）。在这种情况下，盘可能会进入亚开普勒状态，从而降低其内半径处物质的角速度。然而，对通道演化及其在这些情景下对黑洞源性质影响的建模极为复杂，超出了本文的研究范围。

4.2. 发射线

在脉冲星增亮阶段，观测到1.021 keV处Ne Lyβ线的等效宽度（EW）从22.6 eV变化至50.9 eV，其强度也增加了约8倍，这反映了吸积图像向新阶段的转变。这可能表明吸积盘内半径处的等离子体参数发生了变化（Kii等人1986），或表明由开普勒盘与恒星磁场在共转半径处相互作用引起的星风发生了变化（Lovelace等人1995）。如Perna等人（2006）所示，质量外流率可能在自转减速阶段更高，从而导致物质在吸积盘内外半径之间周期性循环。这种反馈会导致吸积盘内质量重新分布，并可能引起系统中质量转移率的波动。若这是2008年扭矩反转后观测到光度增加的原因，则盘内物质径向运动的时间尺度为$\tau_r \sim$ 2年。如果圆盘外半径处的 α 参数相对较小，则可以实现这种情况。

$$\alpha \sim 0.02\, m^{1/2} \left(\frac{c_s}{10^6\ \mathrm{cm\ s^{-1}}}\right)^{-2} \left(\frac{\tau_r}{2\ \mathrm{yr}}\right)^{-1} \left(\frac{R}{10^{10}\ \mathrm{cm}}\right)^{1/2}, \qquad (2)$$

相应地，盘内物质的密度显著高于 X 射线脉冲星的平均值（相关讨论可参见例如King 2012）。此处c_s表示盘在距离R处的声速。盘内物质的缓慢湍流化也可以解释从 4U 1626−67 观测到的扭矩噪声强度较小的现象（Chakrabarty et al.1997）。

在本研究中，遵循Angelini等人（1995）的方法，我们对在4U 1626−67的*Suzaku* X射线光谱中发现的以1 keV为中心的强发射线复合体进行了分解（见图2）。在两次*Suzaku*观测中，最强的谱线是位于~1.025 keV的类氢Ne Lyα线，这与 ASCA 观测结果一致（Angelini等人，1995）。我们在观测中发现的位于~1.48 keV的宽特征，接近于预期的类氢Ne复合边能量1.362 keV。测得的~1.48 keV谱线57 eV的展宽表明，我们可能也探测到了Ne的复合连续谱，正如Angelini等人（1995）在 ASCA 观测中所建议的那样。我们指出，Mg Hα跃迁发生在1.471 keV的能量处（该线未被*Suzaku*分辨），因此这条Mg谱线可能与Ne的复合连续谱发生混合。此外，利用*Suzaku*还探测到了一条中心位于~1.34 keV的第八条谱线，我们暂将其识别为Mg Heα（见表2）。

在自转减速期间，将2006年3月*朱雀*观测中检测到的低能谱线强度与*ASCA*的先前测量结果进行比较，我们发现总体下降幅度约为2.5倍（见表2）。在2008年扭矩反转之前，像*BeppoSAX*（Owens等人1997）、*Chandra*（Schulz等人2001；Krauss等人2007）和 XMM（Krauss等人2007）这样的X射线任务，仅能解析出该复杂谱线中的3或4条。总体而言，这些谱线强度的测量结果与我们的结果一致。在2008年扭矩反转后的新自转加速期间，谱线通量总体上增加了约5倍，但位于1.021 keV的Ne Lyα谱线除外，其增加了约8倍。考虑到光度的这种变化，这一结果并不令人意外。遗憾的是，在之前的自转加速阶段，*爱因斯坦*天文台仅在 1979 年对这一约 1 keV 的复合体进行过一次测量；由于数据质量较低，未能对其进行分辨。与 1990 年扭矩反转相关的整体光度下降幅度约为 3 倍。

in the 0.5−10 keV range (Angelini et al. 1995, and references therein). Krauss et al. (2007) found a decrease of the equivalent widths of the Ne Heα 0.914 keV and Ne Lyα 1.021 keV lines during the spin-down period. The behavior of the EW for the O Heα 0.568 keV and O Lyα 0.653 keV lines was more steady during the same period. With *Suzaku* we found a large increase of the EW of the Ne Lyα 1.021 keV line after the 2008 torque reversal. The EW for this line reached almost the same value measured by ASCA in 1993. For the other lines the increase is less pronounced.

The strong emission-line complex at ∼1 keV has been simply explained as being due to a substantial Ne overabundance (Angelini et al. 1995). Angelini et al. (1995) raised the possibility that 4U 1626–67 is embedded in a shock heated gas from the supernova remmant that created the pulsar, but as they point out, the absence of strong lines from Si, S, and Fe is inconsistent with this idea. Instead, Schulz et al. (2001) proposed that this cool material originated in the accretion disk or the mass donor, so that its composition is then a clue to the nature of the companion. Schulz et al. (2001) detected double-peaked emission lines and strong absorption edges of Ne and O in their *Chandra* HETGS spectrum, thus pointing to the companion in 4U 1626−67 being the chemically fractionated core of a C–O or O–Ne–Mg crystallized white dwarf. Schulz et al. (2001) also concluded that the strong edges are due to absorption in cool, metal-rich material local to the source. Krauss et al. (2007) found that the double-peaked emission lines and the line ratios implied that they are formed in a high-density environment, suggesting that they arise somewhere in the accretion disk. Krauss et al. (2007) also noted that their value for the temperature of the line formation region ($\geq 10^6$ K) is characteristic of the highly ionized, optically thin outer layers of the accretion disk. We note that, in our work, single Gaussian line profiles provided an acceptable fit to the observed line shapes, and physical models like *meka* or *vmeka* did not yield good fits.

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在 0.5_10 keV 范围内（Angelini 等人 1995 及其参考文献）。Krauss 等人（2007）发现，在自转减速期间，Ne Heα 0.914 keV 和 Ne Lyα 1.021 keV 谱线的等值宽度有所减小。而 O Heα 0.568 keV 和 O Lyα 0.653 keV 谱线的等值宽度在同一时期则更为稳定。利用*Suzaku*，我们发现 Ne Lyα 1.021 keV 谱线的等值宽度在 2008 年扭矩反转后大幅增加，该谱线的等值宽度几乎达到了 ASCA 在 1993 年测得的相同数值。对于其他谱线，这种增加则不那么明显。

在~1 keV处观测到的强发射线复合体，被简单地解释为是由于Ne的显著过丰度所致（Angelini等人1995）。Angelini等人（1994）提出了一种可能性，即4U 1626−67嵌入在由产生该脉冲星的超新星遗迹激波加热的气体中；但正如他们所指出的，Si、S和Fe的强谱线缺失与这一观点不符。相反，Schulz等人（2001）提出，这种冷物质起源于吸积盘或质量供体，因此其成分可作为推断伴星性质的线索。Schulz等人（2001）在他们的*Chandra* HETGS 光谱中探测到Ne和O的双峰发射线及强吸收边，从而指出4U 1626_67中的伴星是一个经过化学分馏的C−O或O−Ne−Mg结晶白矮星的核心。Schulz等人（2001）还得出结论，这些强吸收边是由源附近冷且富含金属的物质中的吸收所致。Krauss等人（2007）f研究发现，双峰发射线及其线比表明，这些发射线形成于高密度环境中，从而暗示它们产生于吸积盘内的某个位置。Krauss 等人（2007）还指出，他们得出的发射线形成区温度值（$\geq 10^6$ K）具有吸积盘高度电离且光学薄的外层特征。我们注意到，在本研究中，单高斯线型能够较好地拟合观测到的线形，而像*meka*或*vmeka*这样的物理模型则未能获得良好的拟合效果。

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